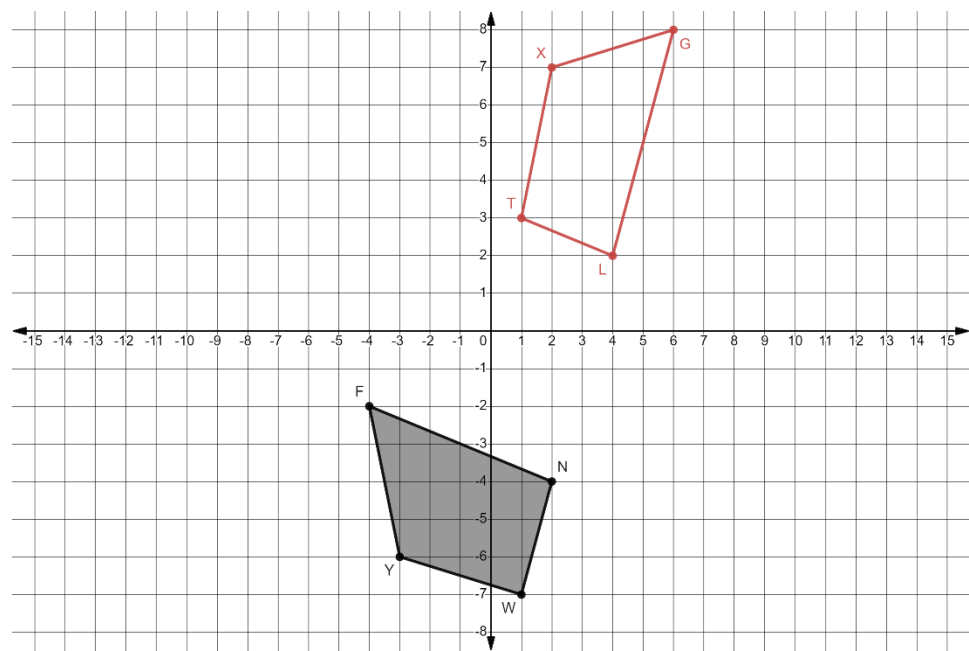


Quadrilateral $FNWY$ is shown on the coordinate grid.



1. Draw the quadrilateral that is the result of a translation left 4 units and up 8 units and then rotated 90° clockwise about the origin. Label the new quadrilateral $GLTX$.

Complete the table.

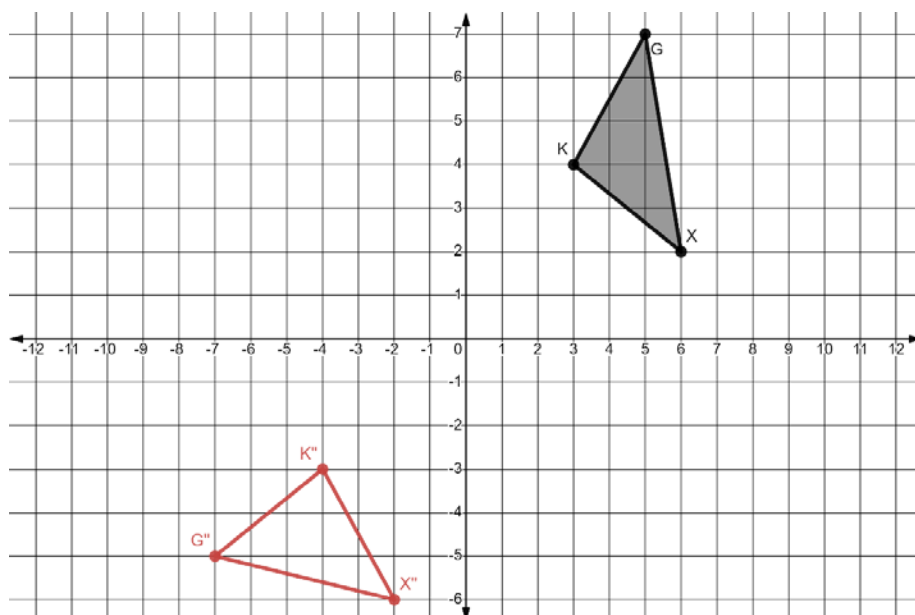
	Transformation	Mapping	Algebraic Description
2.	Translation left 4, up 8	$FNWY \rightarrow F'N'W'Y'$	$(x, y) \rightarrow (x - 4, y + 8)$
3.	Rotation 90° clockwise	$F'N'W'Y' \rightarrow GLTX$	$(x, y) \rightarrow (y, -x)$

4. Complete the table of values for each quadrilateral.

Quadrilateral $FNWY$			Quadrilateral $F'N'W'Y'$			Quadrilateral $GLTX$		
Vertex	x	y	Vertex	x	y	Vertex	x	y
F	-4	-2	F'	-8	6	G	6	8
N	2	-4	N'	-2	4	L	4	2
W	1	-7	W'	-3	1	T	1	3
Y	-3	-6	Y'	-7	2	X	2	7

5. Explain how the tables for quadrilaterals $FNWY$ and $F'N'W'Y'$ show the effect of the transformation on the coordinates that result in the algebraic description.
The translation subtracts 4 from the x -coordinate and adds 8 to the y -coordinate.
6. Explain how the tables for quadrilaterals $F'N'W'Y'$ and $GLTX$ show the effect of the transformation on the coordinates that result in the algebraic description.
The rotation changes the x -coordinate to its opposite, then switches the coordinates.

Triangle XGK is shown on the coordinate grid.



7. Draw the triangle that is the result of a rotation 270° counterclockwise about the origin and then a reflection across the y -axis. Label the new triangle $X''G''K''$.

Complete the table.

	Transformation	Mapping	Algebraic Description
8.	Rotation 270° counterclockwise	$\triangle XGK \rightarrow \triangle X'G'K'$	$(x, y) \rightarrow (y, -x)$
9.	Reflection over y -axis	$\triangle X'G'K' \rightarrow \triangle X''G''K''$	$(x, y) \rightarrow (-x, y)$

10. Describe the connection between the algebraic description of the rotation and the vertices of $\triangle XGK$ and $\triangle X'G'K'$.
The rotation changes the x -coordinate to its opposite, then switches the coordinates.